

Math 63: Real Analysis

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Problem 1. Discuss the continuity of the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational,} \\ \frac{1}{q} & \text{if } x = \frac{p}{q} \text{ with } p, q \in \mathbb{Z} \text{ and } \gcd(p, q) = 1. \end{cases}$$

Problem 2. Let E, E' be metric spaces, $f : E \rightarrow E'$ a continuous function. Prove that if E is compact and f is one-one onto then $f' : E' \rightarrow E$ is continuous. (Hint: f sends closed sets onto closed sets, therefore open sets onto open sets.)

Problem 3. A metric space E is said to be arcwise connected if, given any $p, q \in E$, there is a continuous function $f : [0, 1] \rightarrow E$ such that $f(0) = p, f(1) = q$. Show that any connected open subset of E^n is arcwise connected.

Problem 4. Let $f, f_1, f_2, f_3, \dots$ be continuous real-valued functions on the compact metric space E , with $\lim_{n \rightarrow \infty} f_n$. Prove that if $f_1(p) \leq f_2(p) \leq f_3(p) \leq \dots$ for all $p \in E$ then the sequence $f_1, f_2, f_3, \dots$ converges uniformly.

Problem 5. If E is a compact metric space and $p_0 \in E$ we get a real-valued function F on $C(E)$ by setting $F(f) = f(p_0)$ for all $f \in C(E)$. Prove that F is uniformly continuous.